Project Initialization and Planning Phase

Project Proposal (Proposed Solution) report

Date	3 July 2026
Team ID	
Project Title	**Voice based concept understanding analyser**
Maximum Marks	5 Marks

PROJECT OVERVIEW

Objective	To revolutionize oral evaluations and viva-voce assessments by deploying an automated speech-to-text and machine learning pipeline that analyzes spoken student explanations for absolute conceptual clarity and precision.
Scope	Includes real-time microphonic speech recording, acoustic background filtering, transcription mapping, and high-dimensional semantic vector alignment checks against an academic curriculum baseline.

PROBLEM STATEMENT

Description	Traditional online assessment mechanisms rely heavily on static text or multiple-choice formats, which fail to evaluate verbal expression, track analytical depth, or accurately verify a student's true concept articulation capabilities.
Impact	This limitation creates structural evaluation backlogs, increases human scoring biases during manual viva-voces, and delays feedback loops, resulting in undetected educational comprehension gaps.

PROPOSED SOLUTION

Approach	Build a seamless web framework powered by OpenAI Whisper API for highly adaptive vocal transcription and Sentence-Transformers to map dense semantic vector matches against verified academic rubrics.
Key Features	• Hybrid translation core combining Whisper API audio parsing with automated semantic validation. • Asynchronous FastAPI backend processing system for rapid, low-latency audio packet validation. • Dynamic, custom analytical reporting tracking clear keyword milestones and vocabulary depth.

RESOURCE REQUIREMENTS

RESOURCE TYPE	DESCRIPTION	SPECIFICATION/ALLOCATION
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA L4 Cloud Compute Instances (GCP/AWS)
Memory	RAM specifications	16 GB High-Speed System RAM
Storage	Disk space for data, models, and logs	500 GB NVMe SSD Cloud Block Storage
Software		
Frameworks	Python / Frontend frameworks	Python (FastAPI Backend) & React.js (Frontend UI)
Libraries	Additional libraries	scikit-learn, Transformers, pandas, numpy, soundfile
Development Environment	IDE	VS Code & Jupyter Notebook Environments
Data		
Data Split 1	Source, size, format	LibriSpeech Open Audio Corpus (~10 GB, .wav/.mp3 format)
Data Split 2	Source, size, format	Custom Curated Educational Rubrics Database (.json/.csv structure)